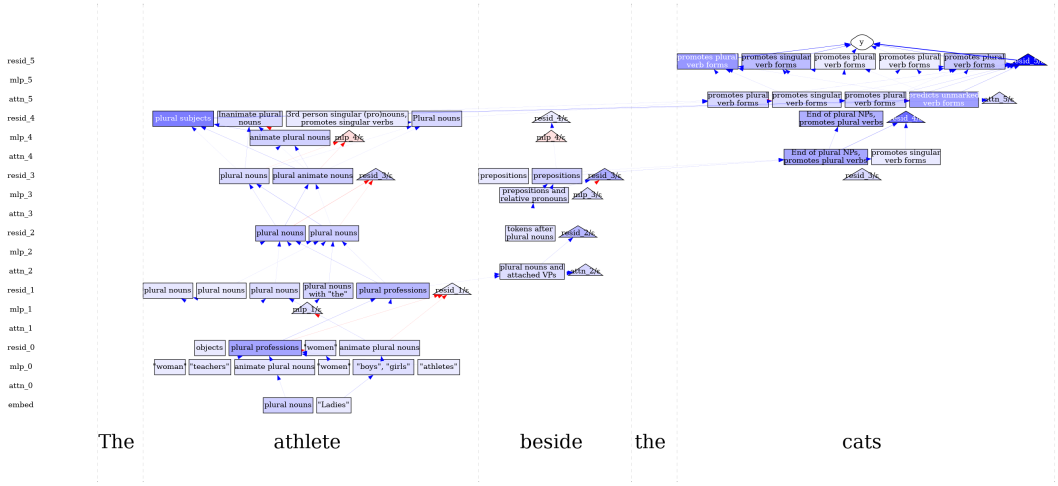

Figure 9: The feature circuit for agreement across a prepositional phrase in Pythia-70M, computed using $T_N = 0.1$ and $T_E = 0.01$. The model detects the subject's number at the subject position. Other features detect prepositional phrases (the start of the distractor clause). Finally, at the last token of the RC, the attention moves the subject information to the last position, where it assists in predicting the correct verb inflection.

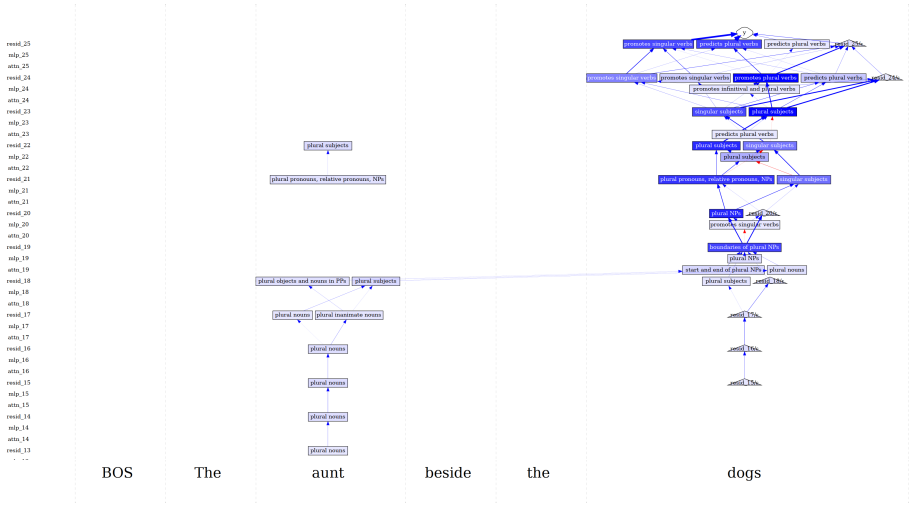

Figure 10: The feature circuit for agreement across a prepositional phrase in Gemma-2-2B, computed using $T_N = 0.5$ and $T_E = 0.05$. Note that we show the circuit beginning in layer 13, as our circuit discovery implicated only one node in earlier layers.

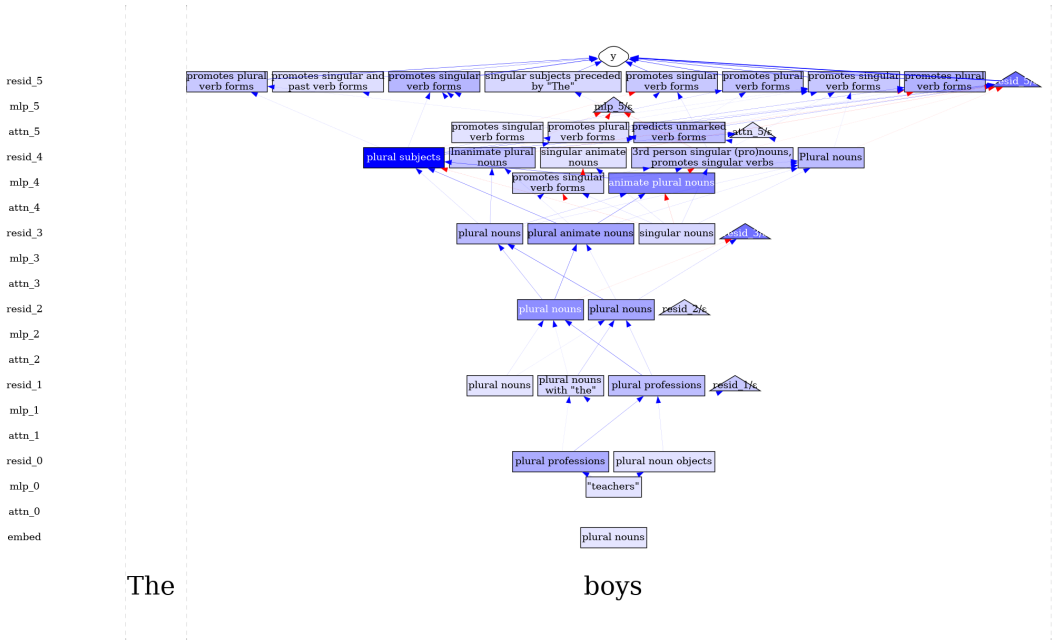

Figure 11: The feature circuit for simple agreement in Pythia-70M, computed using $T_N = 0.2$ and $T_E = 0.02$. The model detects the subject's number at the subject position in early layers. In later layers, these are inputs to features which activate on anything predictive of particular verb inflections.

features in the layer 3 attention seem to be general; they increment many different numbers and letters (as in Figure 5). The induction features are sensitive only to specific tokens: for example, contexts of the form "$x3 \ldots x3$", where "3" is a literal. These compose to form **specific successor** features in layer 5: the most strongly-activating layer 5 residual feature specifically increments "3" to "4" given induction-like lists, where each list item is preceded by the same string (e.g., "Chapter 1…Chapter 2…Chapter 3…Chapter").

The circuit for predicting infinitival objects (Figure 18, discovered with $T_N = 0.25$ and $T_E = 0.001$) contains two distinct mechanisms. First, the model detects the presence of specific verbs like "remember" or "require" which often take infinitival objects. Then, the model uses two separate mechanisms to predict infinitive objects. The first mechanism detects present-tense verbs, participles, or predicate adjectives which can be immediately followed by infinitival direct objects (e.g., "They were excited to…"). The second mechanism detects nominal direct objects that can directly precede infinitival object complements (e.g., "They asked us to…"). Finally, these two mechanisms both influence the output in layer 5 without fully intersecting.

D Sample Features

D.1 Sparse Features

Here, we present examples of sparse features with high indirect effects on the Bias in Bios task. Some of these features clearly activate on terms related to medicine or academia, which are related to the target profession classification task. Others simply detect the presence of "he" or female names.

D.2 Neurons

For contrast, we also present examples of *dense* features—that is, neurons from MLPs, layer-end residuals, and the out-projection of the attention—with high indirect effects on the Bias in Bios task. We cannot directly interpret the activation patterns of these neurons, and so it is difficult to

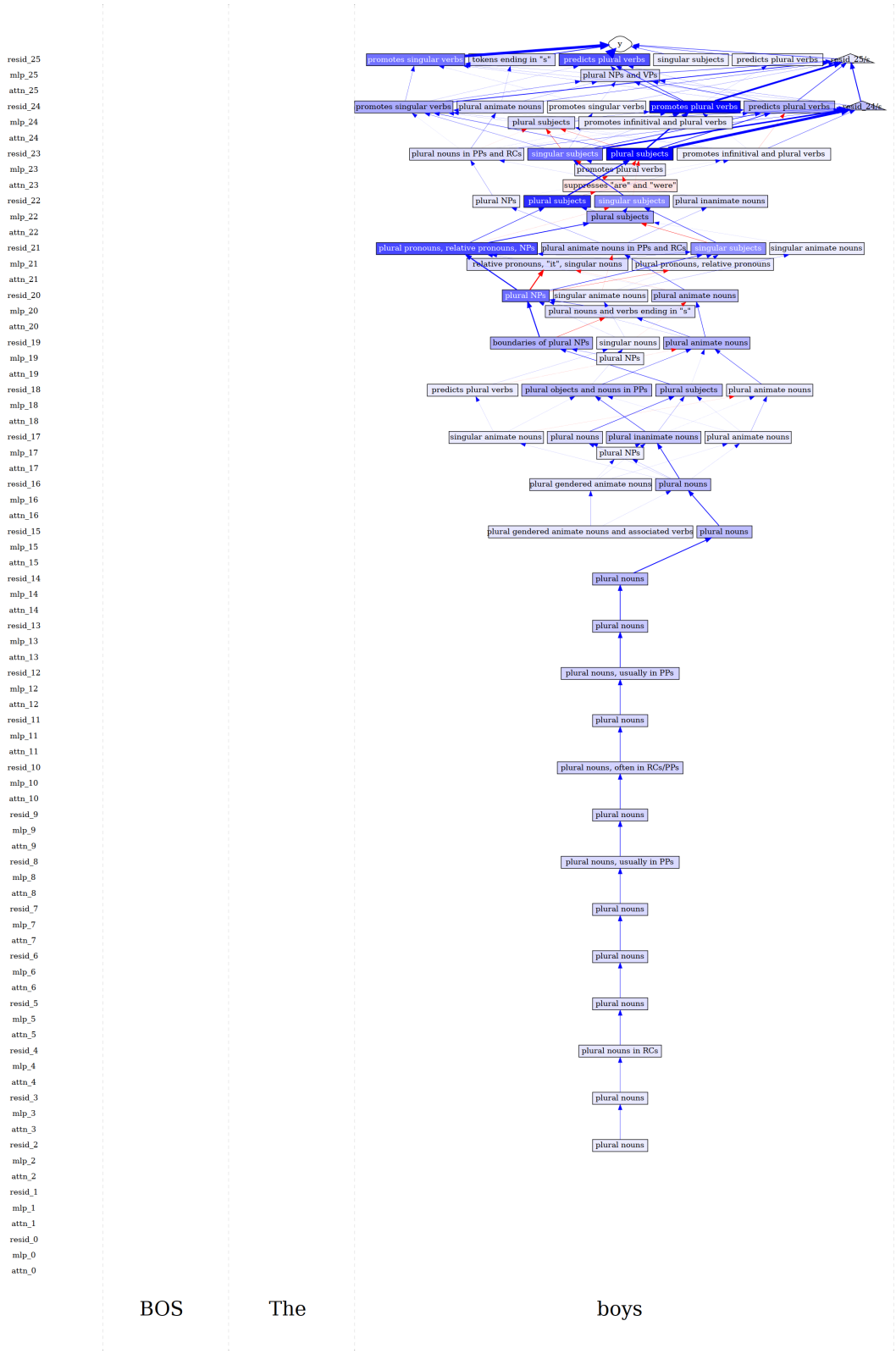

Figure 12: The feature circuit for simple agreement in Gemma-2-2B, computed using $T_N = 0.5$ and $T_E = 0.05$.